Code

Edge Threshold Sensitivity Analysis

Robustness of network topological properties across pruning thresholds

Automated pipeline

2026-04-08

1. Overview

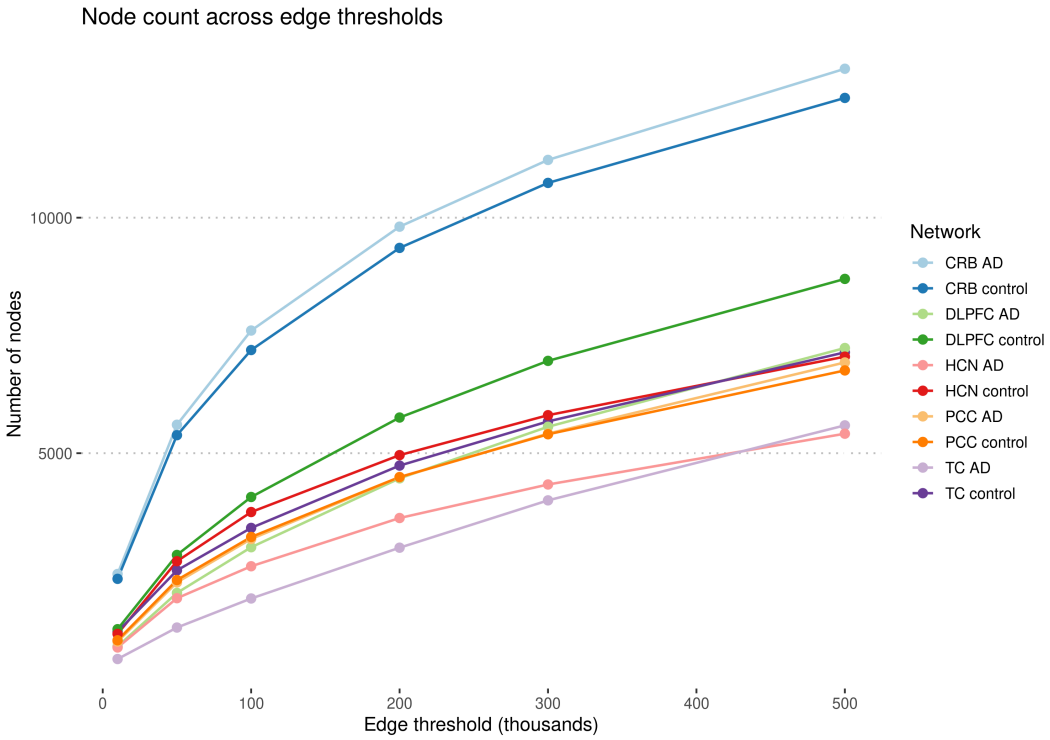
This report evaluates the sensitivity of global network topological properties to the number of edges retained during network pruning. Networks were pruned at thresholds of **10, 50, 100, 200, 300, 500** thousand edges (top-N by mutual information score). If the key findings reported in the manuscript are robust, the **relative ordering** of networks across metrics should remain stable across thresholds, even as absolute values change.

Networks and thresholds analysed

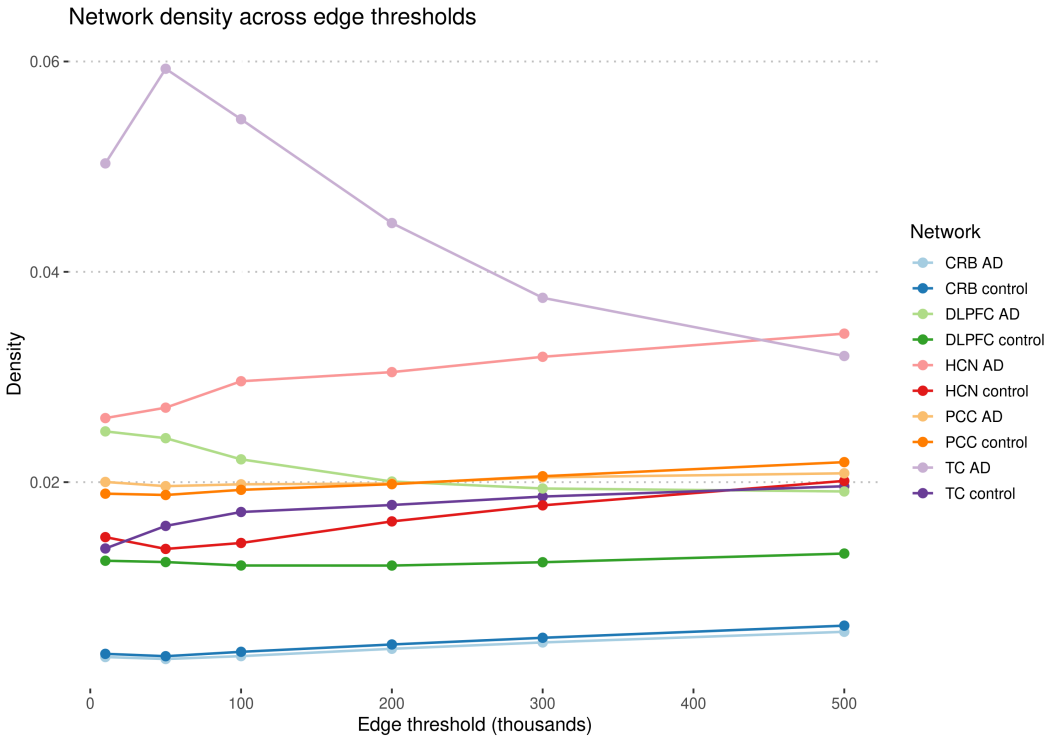
region	phenotype	thresholds_tested
CRB	AD	6
CRB	control	6
DLPFC	AD	6
DLPFC	control	6
HCN	AD	6
HCN	control	6
PCC	AD	6
PCC	control	6
TC	AD	6
TC	control	6

2. Global metrics across thresholds

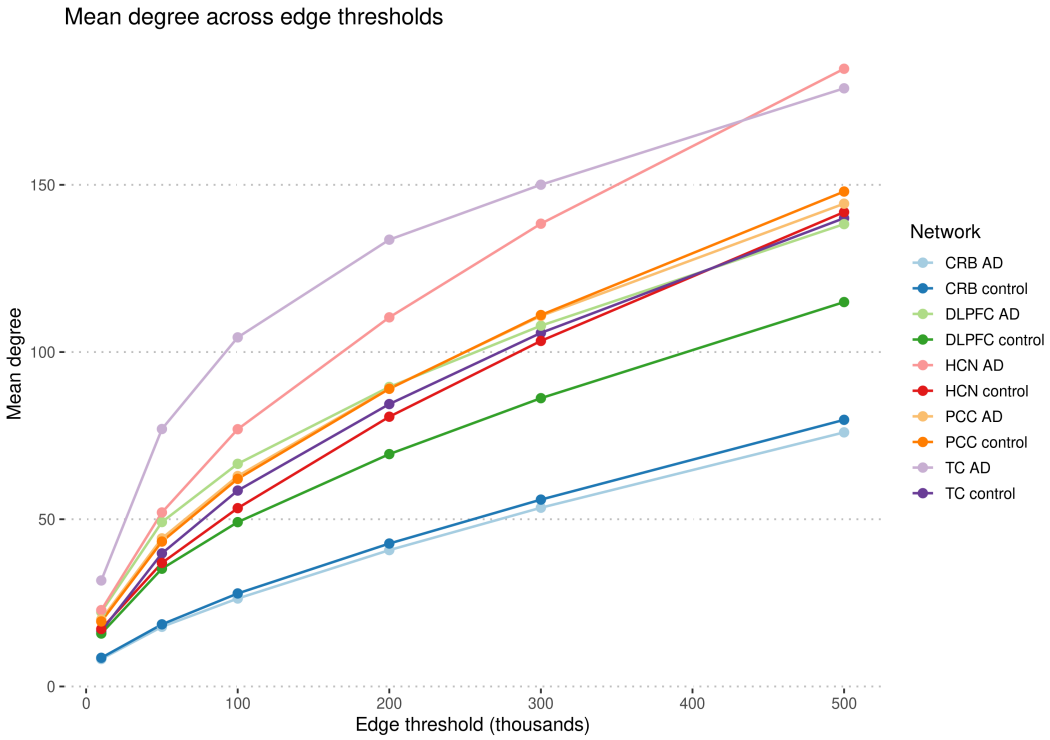
2.1 Number of nodes



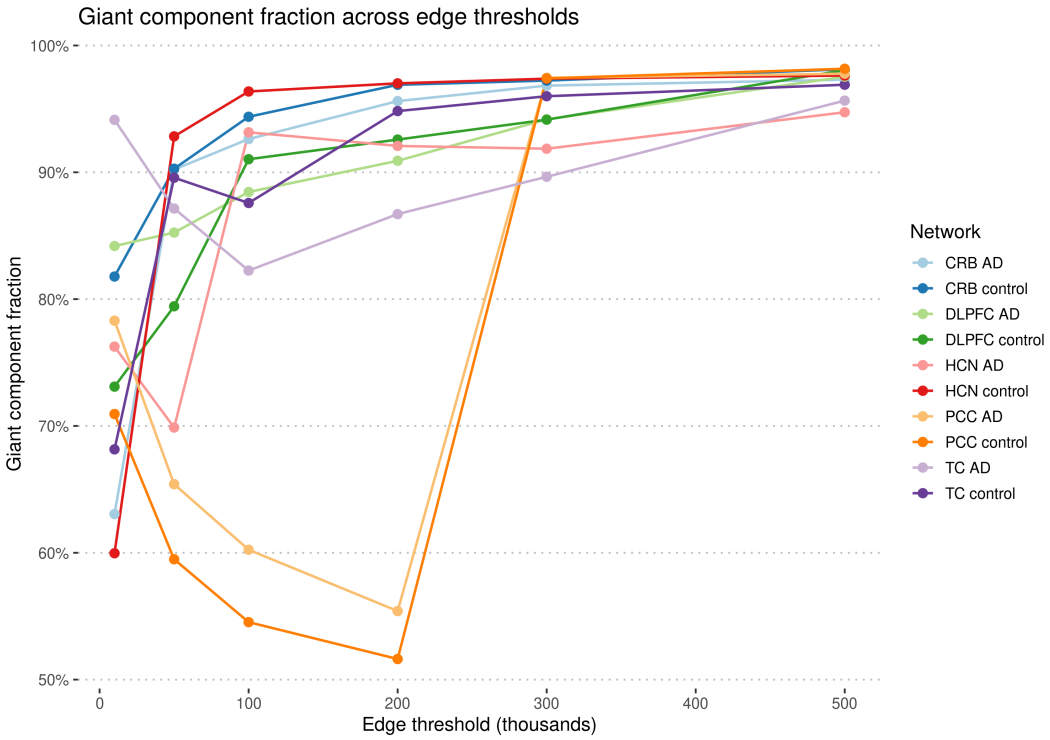
2.2 Network density



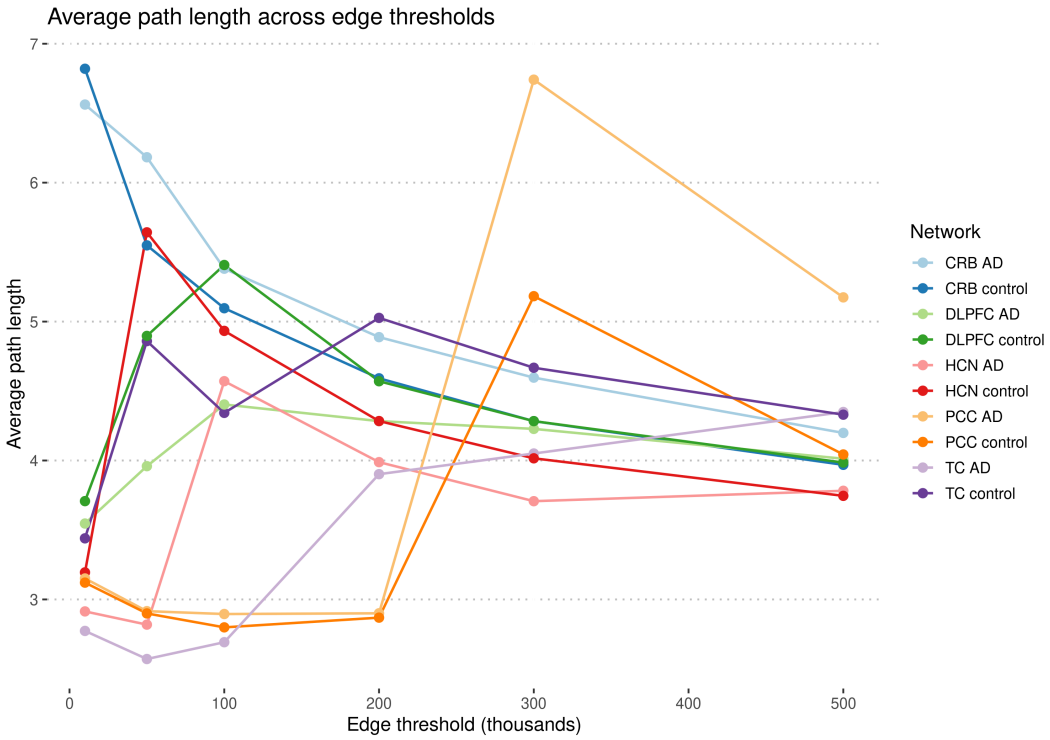
2.3 Mean degree



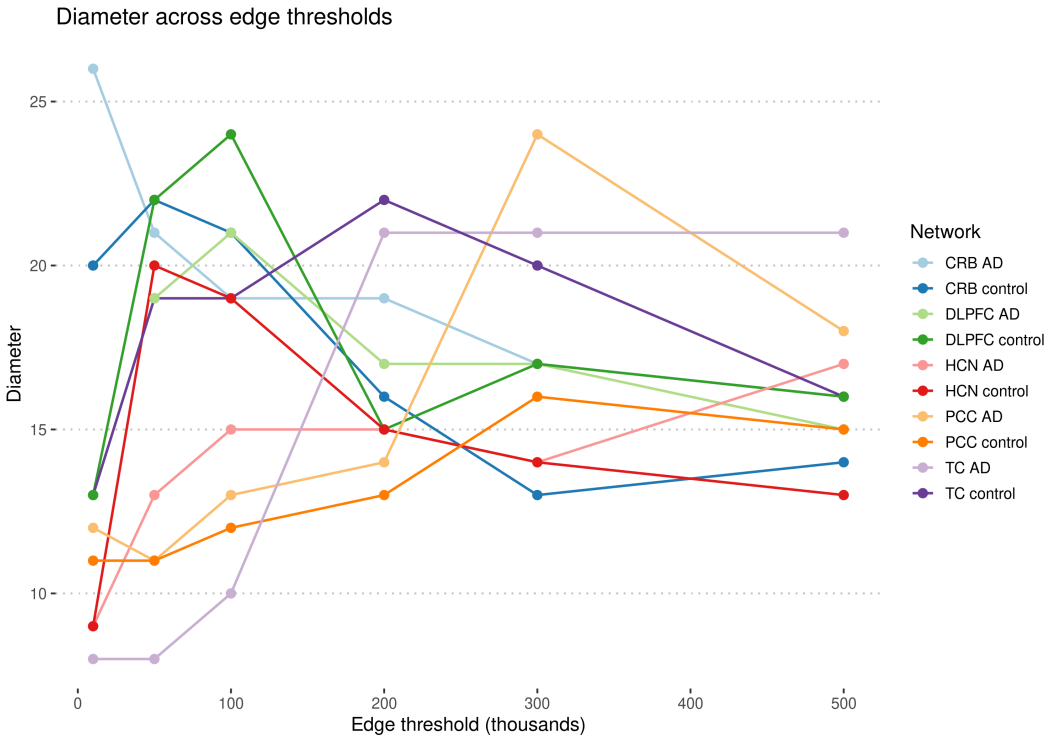
2.4 Giant component fraction



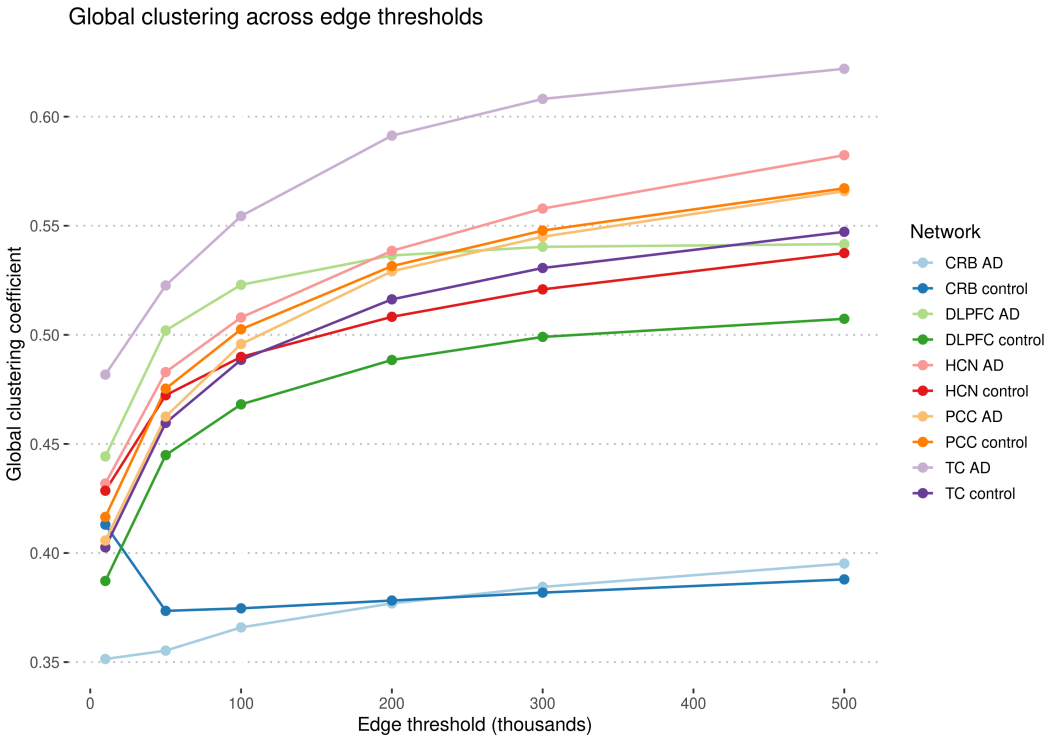
2.5 Average path length



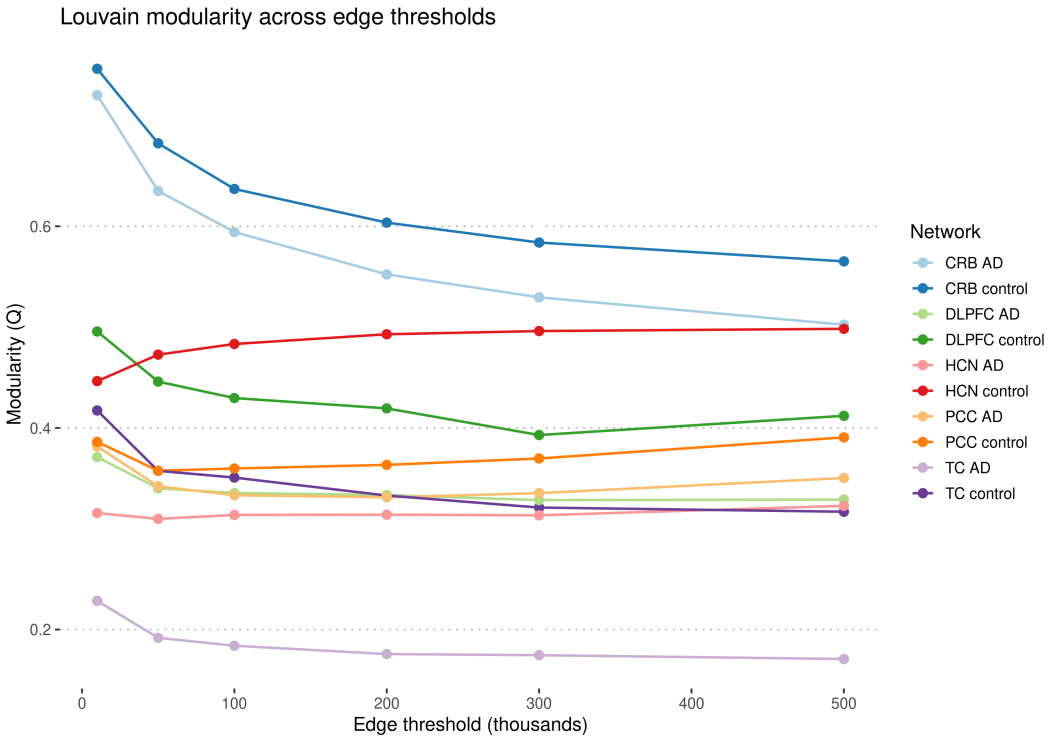
2.6 Diameter



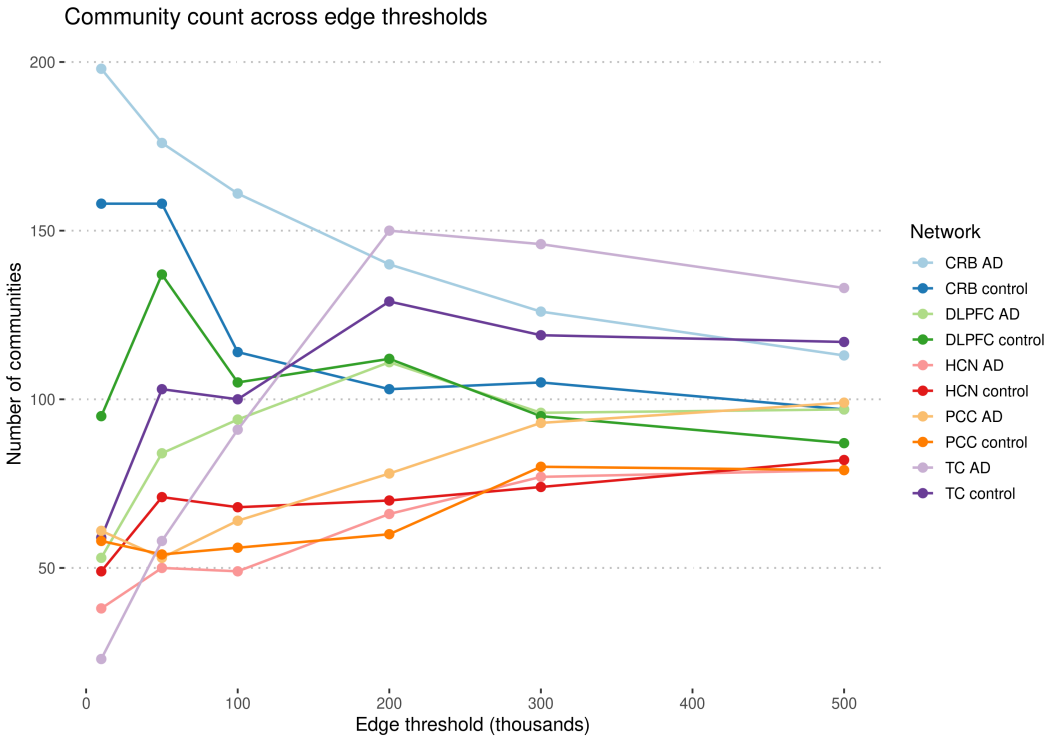
2.7 Clustering coefficient (global)



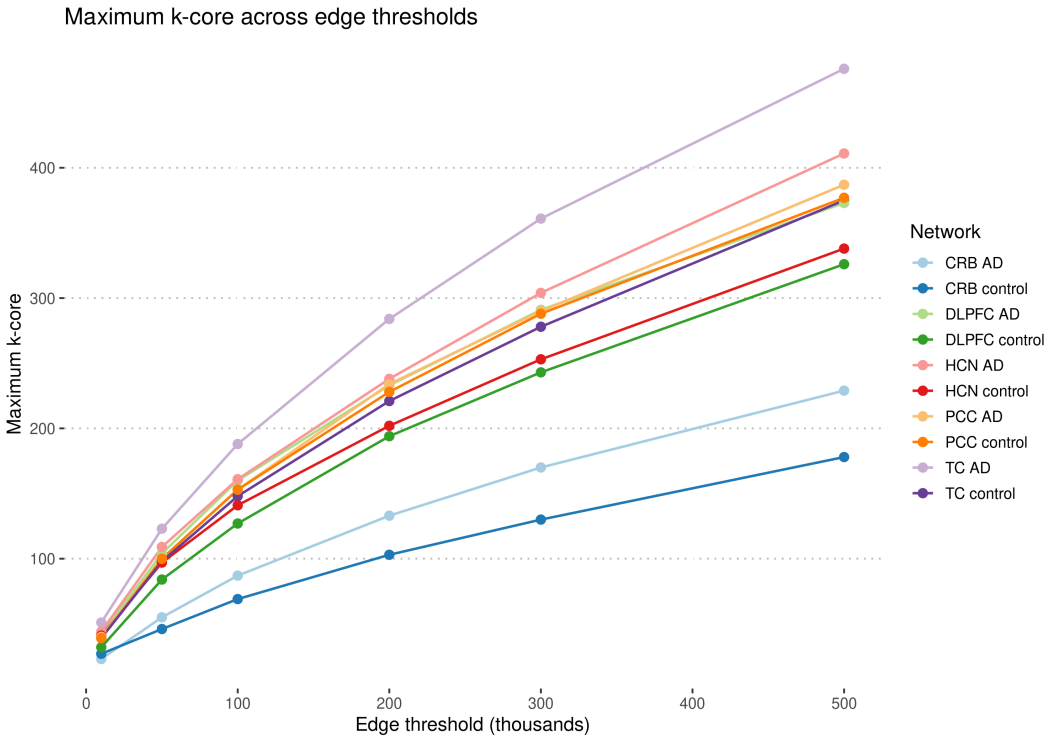
2.8 Modularity (Louvain Q)



2.9 Number of communities



2.10 Max k-core



3. Key robustness tests

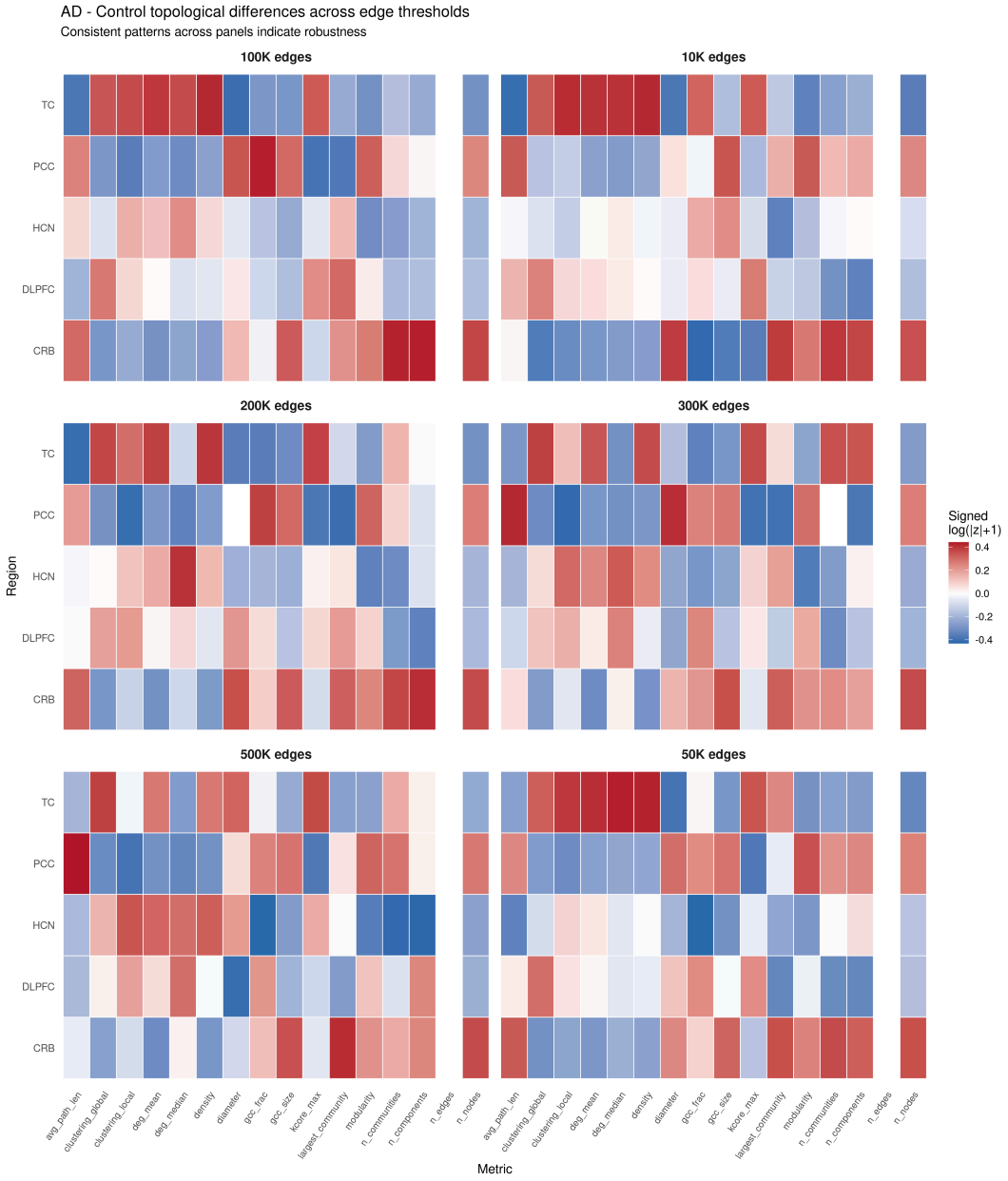
3.1 Regional clustering hierarchy stability (PCA)

For each threshold, we compute a PCA of all 17 topological metrics and plot networks in PC1-PC2 space. If the regional grouping persists across panels, the hierarchy is not an artifact of the 200K cutoff.



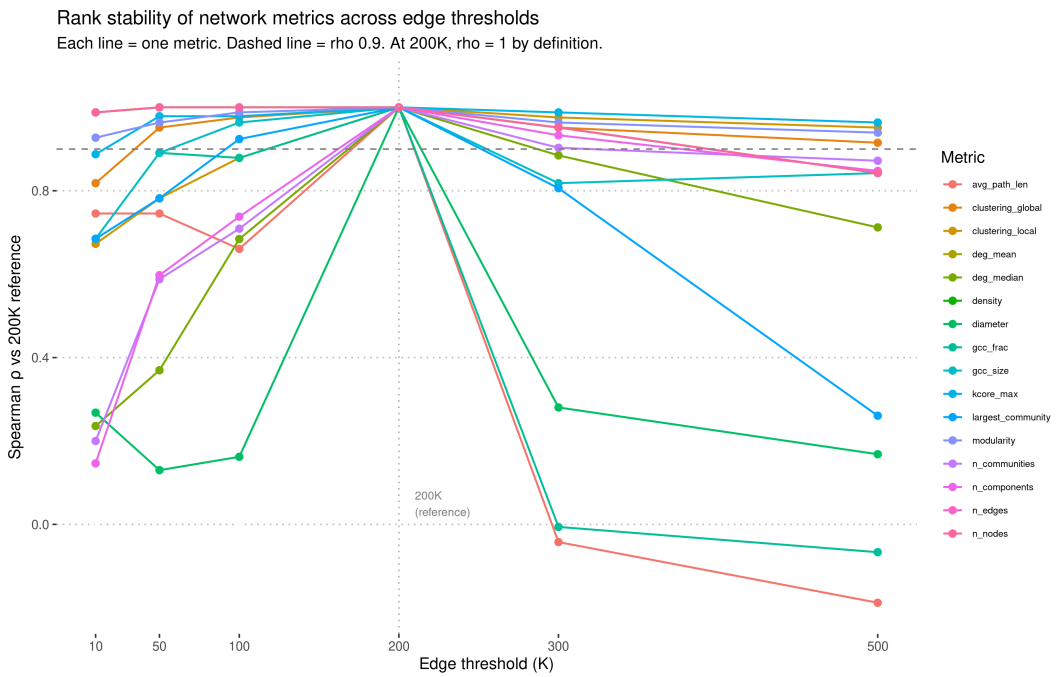
3.2 AD vs Control difference heatmap stability

For each threshold, we compute the signed log-transformed Z-score of AD minus control metric means per region. Consistent colour patterns across thresholds confirm robustness of the contraction-densification finding.



3.3 Rank correlation of metrics across thresholds

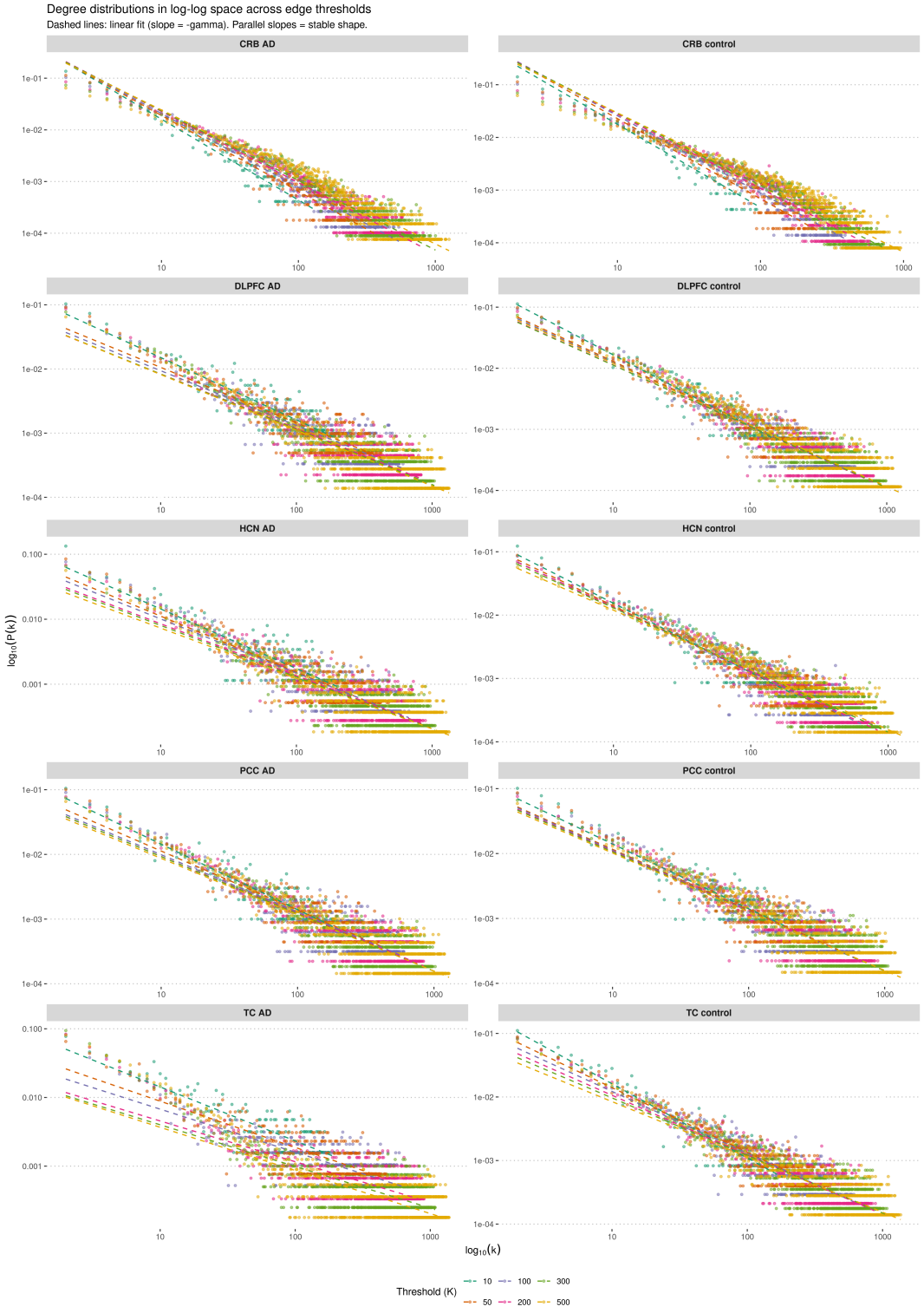
For each metric, we rank all 10 networks at each threshold, then compute the Spearman correlation of that ranking against the ranking at the reference threshold (200K). If \(\rho > 0.9\) across all thresholds, the relative ordering of networks is preserved and the 200K cutoff is not driving the results.



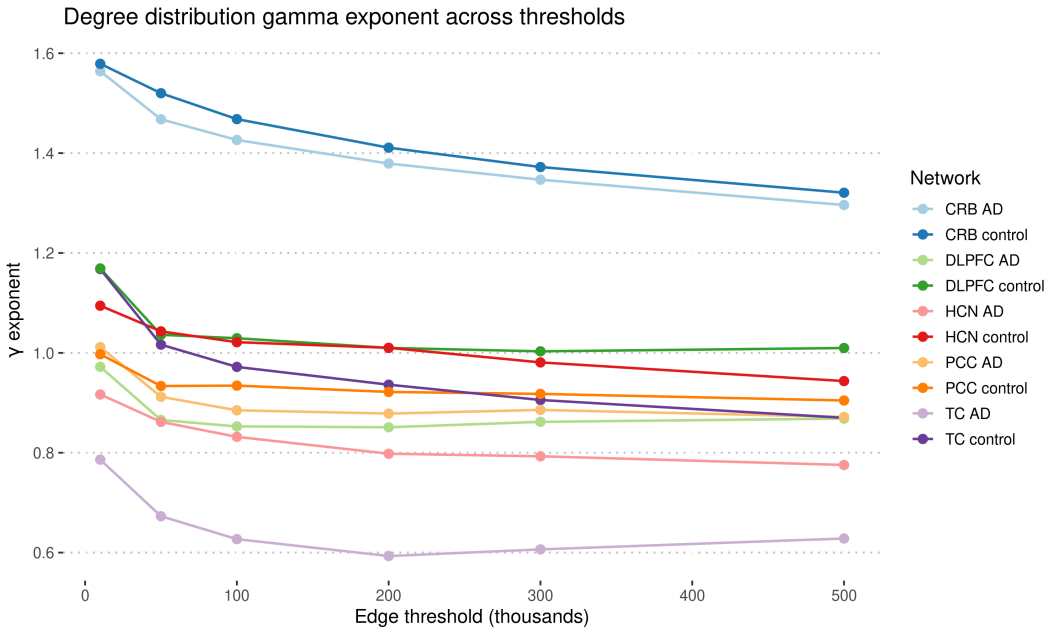
4. Degree distributions

4.1 Log-log degree distributions per network

Each panel shows one network (region + phenotype). Within each panel, each threshold is a separate colour. The dashed line is the linear fit in log-log space whose slope gives the \(\gamma\) exponent shown in the annotation. If the slopes are parallel across thresholds within each panel, the shape of the degree distribution is preserved regardless of the pruning cutoff.


Degree distributions in log-log space across edge thresholds
Dashed lines: linear fit (slope = -gamma). Parallel slopes = stable shape.
CRB AD
CRB control
DLPFC AD
DLPFC control
HCN AD
HCN control
PCC AD
PCC control
TC AD
TC control
log10(P(k))
log10(k)
Threshold (K)
10
50
100
200
300
500

4.2 Gamma exponent across thresholds



4.3 Summary tables

Degree distribution: gamma exponent and best-fit model across thresholds

network	region	phenotype	threshold_k	gamma	best_fit
Mayo_CRB_counts_AD	CRB	AD	10	1.564	Gaussian
Mayo_CRB_counts_AD	CRB	AD	50	1.468	Gaussian
Mayo_CRB_counts_AD	CRB	AD	100	1.426	Gaussian
Mayo_CRB_counts_AD	CRB	AD	200	1.379	Gaussian
Mayo_CRB_counts_AD	CRB	AD	300	1.347	Gaussian
Mayo_CRB_counts_AD	CRB	AD	500	1.296	Gaussian
Mayo_CRB_counts_control	CRB	control	10	1.579	Gaussian
Mayo_CRB_counts_control	CRB	control	50	1.520	Gaussian
Mayo_CRB_counts_control	CRB	control	100	1.468	Gaussian
Mayo_CRB_counts_control	CRB	control	200	1.411	Gaussian
Mayo_CRB_counts_control	CRB	control	300	1.372	Gaussian
Mayo_CRB_counts_control	CRB	control	500	1.321	Gaussian
ROSMAP_DLPFC_counts_AD	DLPFC	AD	10	0.972	Gaussian
ROSMAP_DLPFC_counts_AD	DLPFC	AD	50	0.866	Gaussian
ROSMAP_DLPFC_counts_AD	DLPFC	AD	100	0.853	Gaussian
ROSMAP_DLPFC_counts_AD	DLPFC	AD	200	0.851	Gaussian
ROSMAP_DLPFC_counts_AD	DLPFC	AD	300	0.862	Gaussian
ROSMAP_DLPFC_counts_AD	DLPFC	AD	500	0.868	Gaussian
ROSMAP_DLPFC_counts_control	DLPFC	control	10	1.169	Gaussian
ROSMAP_DLPFC_counts_control	DLPFC	control	50	1.036	Gaussian

network	region	phenotype	threshold_k	gamma	best_fit
ROSMAP_DLPFC_counts_control	DLPFC	control	100	1.029	Gaussian
ROSMAP_DLPFC_counts_control	DLPFC	control	200	1.010	Gaussian
ROSMAP_DLPFC_counts_control	DLPFC	control	300	1.003	Gaussian
ROSMAP_DLPFC_counts_control	DLPFC	control	500	1.010	Gaussian
ROSMAP_HCN_counts_AD	HCN	AD	10	0.917	Exponential
ROSMAP_HCN_counts_AD	HCN	AD	50	0.862	Gaussian
ROSMAP_HCN_counts_AD	HCN	AD	100	0.832	Gaussian
ROSMAP_HCN_counts_AD	HCN	AD	200	0.798	Gaussian
ROSMAP_HCN_counts_AD	HCN	AD	300	0.793	Gaussian
ROSMAP_HCN_counts_AD	HCN	AD	500	0.775	Gaussian
ROSMAP_HCN_counts_control	HCN	control	10	1.094	Gaussian
ROSMAP_HCN_counts_control	HCN	control	50	1.043	Gaussian
ROSMAP_HCN_counts_control	HCN	control	100	1.021	Gaussian
ROSMAP_HCN_counts_control	HCN	control	200	1.010	Gaussian
ROSMAP_HCN_counts_control	HCN	control	300	0.981	Gaussian
ROSMAP_HCN_counts_control	HCN	control	500	0.943	Gaussian
ROSMAP_PCC_counts_AD	PCC	AD	10	1.012	Gaussian
ROSMAP_PCC_counts_AD	PCC	AD	50	0.912	Gaussian
ROSMAP_PCC_counts_AD	PCC	AD	100	0.885	Gaussian
ROSMAP_PCC_counts_AD	PCC	AD	200	0.878	Gaussian
ROSMAP_PCC_counts_AD	PCC	AD	300	0.886	Gaussian
ROSMAP_PCC_counts_AD	PCC	AD	500	0.872	Gaussian
ROSMAP_PCC_counts_control	PCC	control	10	0.997	Gaussian
ROSMAP_PCC_counts_control	PCC	control	50	0.934	Gaussian
ROSMAP_PCC_counts_control	PCC	control	100	0.934	Gaussian
ROSMAP_PCC_counts_control	PCC	control	200	0.922	Gaussian
ROSMAP_PCC_counts_control	PCC	control	300	0.918	Gaussian
ROSMAP_PCC_counts_control	PCC	control	500	0.905	Gaussian
Mayo_TC_counts_AD	TC	AD	10	0.786	Gaussian
Mayo_TC_counts_AD	TC	AD	50	0.673	Gaussian
Mayo_TC_counts_AD	TC	AD	100	0.627	Gaussian
Mayo_TC_counts_AD	TC	AD	200	0.593	Gaussian
Mayo_TC_counts_AD	TC	AD	300	0.606	Gaussian
Mayo_TC_counts_AD	TC	AD	500	0.628	Gaussian
Mayo_TC_counts_control	TC	control	10	1.168	Gaussian
Mayo_TC_counts_control	TC	control	50	1.016	Gaussian
Mayo_TC_counts_control	TC	control	100	0.972	Gaussian
Mayo_TC_counts_control	TC	control	200	0.936	Gaussian
Mayo_TC_counts_control	TC	control	300	0.906	Gaussian
Mayo_TC_counts_control	TC	control	500	0.870	Gaussian

Best-fit degree distribution model per network across thresholds

region	phenotype	thr_10	thr_100	thr_200	thr_300	thr_50	thr_500
CRB	AD	Gaussian	Gaussian	Gaussian	Gaussian	Gaussian	Gaussian
CRB	control	Gaussian	Gaussian	Gaussian	Gaussian	Gaussian	Gaussian
DLPFC	AD	Gaussian	Gaussian	Gaussian	Gaussian	Gaussian	Gaussian
DLPFC	control	Gaussian	Gaussian	Gaussian	Gaussian	Gaussian	Gaussian
HCN	AD	Exponential	Gaussian	Gaussian	Gaussian	Gaussian	Gaussian
HCN	control	Gaussian	Gaussian	Gaussian	Gaussian	Gaussian	Gaussian
PCC	AD	Gaussian	Gaussian	Gaussian	Gaussian	Gaussian	Gaussian
PCC	control	Gaussian	Gaussian	Gaussian	Gaussian	Gaussian	Gaussian
TC	AD	Gaussian	Gaussian	Gaussian	Gaussian	Gaussian	Gaussian
TC	control	Gaussian	Gaussian	Gaussian	Gaussian	Gaussian	Gaussian

5. Interpretation guide

What each section answers for the reviewer:

1. **Section 2 (line plots):** If lines run roughly parallel without crossing, the relative differences between networks are preserved regardless of the edge threshold. Vertical spread at each threshold shows the magnitude of inter-network differences; what matters is that the *ordering* is stable.

2. **Section 3.1 (PCA facets):** If regional clusters (same colour) stay together across panels, the finding that networks group by region rather than disease status is robust.

3. **Section 3.2 (heatmaps):** If the colour pattern (which regions show increases vs decreases in which metrics) is consistent across panels, the contraction-densification pattern and CRB expansion pattern are not artifacts.

4. **Section 3.3 (rank correlations):** Spearman rho = 1 at 200K by definition (reference). If rho stays above 0.9 at other thresholds, the 200K cutoff produces the same relative ordering as any other cutoff. Metrics that drop below 0.9 at extreme thresholds (10K) identify the least robust topological features.

5. **Section 4 (degree distributions):** If all networks are Gaussian at all thresholds, it means the degree distributions are unimodal and symmetric rather than heavy-tailed. This is expected for pruned MI networks because retaining only the top-N edges compresses the upper tail of the degree distribution — the pruning itself imposes a ceiling on maximum degree. The key question is whether the *gamma exponent* is stable, not whether the best-fit model changes.

```
## R version 4.4.1 (2024-06-14)
## Platform: x86_64-pc-linux-gnu
## Running under: Ubuntu 20.04.6 LTS
##
## Matrix products: default
## BLAS:   /usr/lib/x86_64-linux-gnu/blas/libblas.so.3.9.0
## LAPACK: /usr/lib/x86_64-linux-gnu/lapack/liblapack.so.3.
9.0
##
## locale:
##  [1] LC_CTYPE=es_MX.UTF-8       LC_NUMERIC=C
##  [3] LC_TIME=es_MX.UTF-8        LC_COLLATE=es_MX.UTF-8
##  [5] LC_MONETARY=es_MX.UTF-8    LC_MESSAGES=es_MX.UTF-8
##  [7] LC_PAPER=es_MX.UTF-8       LC_NAME=C
##  [9] LC_ADDRESS=C               LC_TELEPHONE=C
## [11] LC_MEASUREMENT=es_MX.UTF-8 LC_IDENTIFICATION=C
##
## time zone: America/Mexico_City
## tzcode source: system (glibc)
##
## attached base packages:
##  [1] splines   stats4    tools     stats     graphics  gr
Devices utils
##  [8] datasets  methods   base
##
## other attached packages:
##  [1] RColorBrewer_1.1-3  ggrepel_0.9.5       knitr_1.51
##  [4] rmarkdown_2.29      poweRlaw_1.0.0      VGAM_1.1-14
##  [7] fitdistrplus_1.1-11 survival_3.5-8      MASS_7.3-65
## [10] optparse_1.7.5      ggsci_3.0.3         ggpubr_0.6.0
## [13] scales_1.4.0        purrr_1.0.2         stringr_1.5.
1
## [16] tidyr_1.3.1         dplyr_1.1.4         ggplot2_4.0.
1
## [19] future.apply_1.11.2 future_1.33.2       igraph_2.2.1
## [22] data.table_1.18.2.1
##
## loaded via a namespace (and not attached):
##  [1] gtable_0.3.6      bslib_0.10.0      xfun_0.56
rstatix_0.7.2
##  [5] lattice_0.22-7    vctrs_0.7.1       generics_0.1.4
parallel_4.4.1
##  [9] getopt_1.20.4     tibble_3.3.1      pacman_0.5.1
pkgconfig_2.0.3
## [13] Matrix_1.6-5      S7_0.2.1          lifecycle_1.0.5
compiler_4.4.1
```

```
## [17] farver_2.1.2      codetools_0.2-20  carData_3.0-5
sass_0.4.10
## [21] htmltools_0.5.9   yaml_2.3.12       pillar_1.11.1
car_3.1-2
## [25] jquerylib_0.1.4   cachem_1.1.0      abind_1.4-8
nlme_3.1-164
## [29] parallelly_1.46.1 tidyselect_1.2.1  digest_0.6.39
stringi_1.8.7
## [33] listenv_0.10.0    labeling_0.4.3    fastmap_1.2.0
grid_4.4.1
## [37] cli_3.6.5         magrittr_2.0.4    broom_1.0.5
withr_3.0.2
## [41] backports_1.5.0   globals_0.18.0    otel_0.2.0
ggsignif_0.6.4
## [45] evaluate_1.0.5    mgcv_1.9-1        rlang_1.1.7
Rcpp_1.1.1
## [49] glue_1.8.0        jsonlite_2.0.0    R6_2.6.1
```